

Derivations

- Derivation 1

$$\begin{aligned}
\rho(E[Y|X], h(x)) &= \sqrt{\frac{\text{Var}[E[Y|X]]}{\text{Var}[Y]}} = \frac{\text{Cov}[E[Y|X], h(x)]}{\sqrt{\text{Var}[E[Y|X]]\text{Var}[Y]}} \sqrt{\frac{\text{Var}[E[Y|X]]}{\text{Var}[Y]}} \\
&= \frac{\text{Cov}[E[Y|X], h(x)]}{\sqrt{\text{Var}[Y]\text{Var}[h(x)]}} \\
&= \frac{E[E[Y|X]h(x)] - E[E[Y|X]]E[h(x)]}{\sqrt{\text{Var}[Y]\text{Var}[h(x)]}} \\
&= \frac{E[E[Y * h(x)|X]] - E[E[Y|X]]E[h(x)]}{\sqrt{\text{Var}[Y]\text{Var}[h(x)]}} \\
&= \text{/law of total expectation/} \\
&= \frac{E[Y * h(x)] - E[Y]E[h(x)]}{\sqrt{\text{Var}[Y]\text{Var}[h(x)]}} \\
&= \frac{\text{Cov}[Y, h(x)]}{\sqrt{\text{Var}[Y]\text{Var}[h(x)]}} \\
&= \rho(Y, h(x))
\end{aligned}$$

- Derivation 2

$$\begin{aligned}
\rho(Y, \alpha + \beta E[Y|X]) &= \frac{\text{Cov}[Y, \alpha + \beta E[Y|X]]}{\sqrt{\text{Var}[E[Y|X]]\text{Var}[Y]}} \sqrt{\frac{\text{Var}[E[Y|X]]}{\text{Var}[Y]}} \\
&= \frac{E[Y(\alpha + \beta E[Y|X])] - E[Y]E[\alpha + \beta E[Y|X]]}{\beta \sqrt{\text{Var}[Y]\text{Var}[E[Y|X]]}} \\
&= \frac{E[Y * E[Y|X]] + E[Y]E[E[Y|X]]}{\sqrt{\text{Var}[Y]\text{Var}[E[Y|X]]}} \\
&= \frac{\text{Cov}[Y, E[Y|X]]}{\sqrt{\text{Var}[Y]\text{Var}[E[Y|X]]}} \\
&= \rho(Y, E[Y|X])
\end{aligned}$$

- Derivation 3

$$\begin{aligned}
\rho_c(V, W) &= 1 - \frac{E[(V - W)^2]}{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2} \\
&= \frac{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2 - E[(V - W)^2]}{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2} \\
&= \frac{2(E[VW] - E[V]E[W])}{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2} \\
&= \frac{2\text{Cov}[V, W]}{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2} \\
&= \frac{2\rho(V, W)\sqrt{\text{Var}[V]\text{Var}[W]}}{\text{Var}[V] + \text{Var}[W] + (E[V] - E[W])^2}
\end{aligned}$$

- Derivation 4

$$h^*(x) = \beta_0 + \beta_1 x$$

$$\text{Minimise } E[(Y - (\beta_0 + \beta_1 X))^2]$$

$$\frac{\partial}{\partial \beta_0} E[(Y - (\beta_0 + \beta_1 X))^2] = 0$$

$$E \left[\frac{\partial}{\partial \beta_0} (Y - (\beta_0 + \beta_1 X))^2 \right] = 0$$

$$E[-2(Y - (\beta_0 + \beta_1 X))] = 0$$

$$E[Y - \beta_0 - \beta_1 X] = 0$$

$$E[Y] - \beta_0 - \beta_1 E[X] = 0$$

$$E[Y] - \beta_1 E[X] = \beta_0$$

$$\frac{\partial}{\partial \beta_1} E[(Y - (\beta_0 + \beta_1 X))^2] = 0$$

$$E \left[\frac{\partial}{\partial \beta_1} (Y - E[Y] - \beta_1(X - E[X]))^2 \right] = 0$$

$$E[-2(X - E[X])(Y - E[Y] - \beta_1(X - E[X]))] = 0$$

$$E[(X - E[X])(Y - E[Y])] - \beta_1 E[(X - E[X])^2] = 0$$

$$E[XY] - E[X]E[Y] - \beta_1 \text{Var}[X] = 0$$

$$\text{Cov}[Y, X] - \beta_1 \text{Var}[X] = 0$$

$$\beta_1 = \frac{\text{Cov}[Y, X]}{\text{Var}[X]}$$

$$\Rightarrow h^*(x_0) = \left(E[Y] - \frac{\text{Cov}[Y, X]}{\text{Var}[X]} E[X] \right) + \frac{\text{Cov}[Y, X]}{\text{Var}[X]} x_0$$

- Derivation 5

$$\sqrt{\frac{\text{Var}[E[Y|X]]}{\text{Var}[Y]}} = / E[Y|X] = \beta_0 + \beta_1 X /$$

$$= \sqrt{\frac{\text{Var}[\beta_0 + \beta_1 X]}{\text{Var}[Y]}}$$

$$= \beta_1 \sqrt{\frac{\text{Var}[X]}{\text{Var}[Y]}}$$

$$\Rightarrow \beta_1 \sqrt{\frac{\text{Var}[X]}{\text{Var}[Y]}} = \frac{\text{Cov}(Y, X)}{\sqrt{\text{Var}[Y]\text{Var}[X]}}$$

$$\begin{aligned}
\frac{\text{Cov}(Y, X)}{\sqrt{\text{Var}[Y]\text{Var}[X]}} &= \frac{\text{E}[YX] - \text{E}[Y]\text{E}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= / \text{ law of total expectation } / \\
&= \frac{\text{E}[\text{E}(YX | X)] - \text{E}[\text{E}[Y|X]]\text{E}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\text{E}[X \text{E}[Y|X]] - \text{E}[\text{E}[Y|X]]\text{E}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= / \text{E}[Y|X] = \beta_0 + \beta_1 X / \\
&= \frac{\text{E}[X(\beta_0 + \beta_1 X)] - \text{E}[\beta_0 + \beta_1 X] \text{E}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\text{E}[\beta_0 X + \beta_1 X^2] - (\beta_0 + \beta_1 \text{E}[X])\text{E}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\beta_0 \text{E}[X] + \beta_1 \text{E}[X^2] - \beta_0 \text{E}[X] - \beta_1 \text{E}[X]^2}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\beta_1 \text{E}[X^2] - \beta_1 \text{E}[X]^2}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\beta_1 (\text{E}[X^2] - \text{E}[X]^2)}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \frac{\beta_1 \text{Var}[X]}{\sqrt{\text{Var}[Y]\text{Var}[X]}} \\
&= \beta_1 \sqrt{\frac{\text{Var}[X]}{\text{Var}[Y]}}
\end{aligned}$$

- Derivation 6

$$h^{**}(X) = \beta_0 + \beta_1 X$$

$$\text{Maximise } \rho_c(Y, \beta_0 + \beta_1 X) = \frac{2 \text{Cov}[Y, \beta_0 + \beta_1 X]}{\text{Var}[Y] \text{Var}[\beta_0 + \beta_1 X] + (\text{E}[Y] - \text{E}[\beta_0 + \beta_1 X])^2}$$

$$\begin{aligned}
\frac{\partial}{\partial \beta_0} \left(\frac{2 \text{Cov}[Y, \beta_0 + \beta_1 X]}{\text{Var}[Y] \text{Var}[\beta_0 + \beta_1 X] + (\text{E}[Y] - \text{E}[\beta_0 + \beta_1 X])^2} \right) &= 0 \\
\frac{\partial}{\partial \beta_0} \left(\frac{2\beta_1 \text{Cov}[Y, X]}{\text{Var}[Y] \beta_1^2 \text{Var}[X] + (\text{E}[Y] - \beta_0 - \beta_1 \text{E}[X])^2} \right) &= 0 \\
\frac{2\beta_1 \text{Cov}[Y, X] \left[\frac{\partial}{\partial \beta_0} \left(\text{Var}[Y] \beta_1^2 \text{Var}[X] + (\text{E}[Y] - \beta_0 - \beta_1 \text{E}[X])^2 \right) \right]}{\left(\text{Var}[Y] \beta_1^2 \text{Var}[X] + (\text{E}[Y] - \beta_0 - \beta_1 \text{E}[X])^2 \right)^2} &= 0 \\
\Rightarrow 2\beta_1 \text{Cov}[Y, X] (-2(\text{E}[Y] - \beta_0 - \beta_1 \text{E}[X])) &= 0 \\
\Rightarrow \text{E}[Y] - \beta_0 - \beta_1 \text{E}[X] &= 0 \\
\Rightarrow \beta_0 &= \text{E}[Y] - \beta_1 \text{E}[X]
\end{aligned}$$

$$\frac{\partial}{\partial \beta_1} \left(\frac{2 \text{Cov}[Y, \beta_0 + \beta_1 X]}{\text{Var}[Y] \text{Var}[\beta_0 + \beta_1 X] + (\text{E}[Y] - \text{E}[\beta_0 + \beta_1 X])^2} \right) = 0$$

$$\frac{\partial}{\partial \beta_1} \left(\frac{2\beta_1 \text{Cov}[Y, X]}{\text{Var}[Y] + \beta_1^2 \text{Var}[X] + (\text{E}[Y] - \beta_0 - \beta_1 \text{E}[X])^2} \right) = 0 \quad \Leftrightarrow \quad \text{insert } \beta_0 = \text{E}[Y] - \beta_1 \text{E}[X]$$

$$\frac{\partial}{\partial \beta_1} \left(\frac{2\beta_1 \text{Cov}[Y, X]}{\text{Var}[Y] + \beta_1^2 \text{Var}[X] + (\text{E}[Y] - \text{E}[Y] + \beta_1 \text{E}[X] - \beta_1 \text{E}[X])^2} \right) = 0$$

$$\frac{\partial}{\partial \beta_1} \left(\frac{2\beta_1 \text{Cov}[Y, X]}{\text{Var}[Y] + \beta_1^2 \text{Var}[X]} \right) = 0$$

$$\frac{2 \text{Cov}[Y, X] (\text{Var}[Y] + \beta_1^2 \text{Var}[X]) - 2\beta_1 \text{Cov}[Y, X] (2\beta_1 \text{Var}[X])}{(\text{Var}[Y] + \beta_1^2 \text{Var}[X])^2} = 0$$

$$\Rightarrow 2 \text{Cov}[Y, X] \text{Var}[Y] + 2\beta_1^2 \text{Cov}[Y, X] \text{Var}[X] - 4\beta_1^2 \text{Cov}[Y, X] \text{Var}[X] = 0$$

$$\Rightarrow 2 \text{Cov}[Y, X] \text{Var}[Y] - 2\beta_1^2 \text{Cov}[Y, X] \text{Var}[X] = 0$$

$$\Rightarrow \text{Var}[Y] - \beta_1^2 \text{Var}[X] = 0$$

$$\Rightarrow \beta_1^2 = \frac{\text{Var}[Y]}{\text{Var}[X]}$$

$$\Rightarrow \beta_1 = \pm \sqrt{\frac{\text{Var}[Y]}{\text{Var}[X]}} \quad \leftarrow \text{Must be positive}$$

$$\Rightarrow h^{**}(x_0) = \left(\text{E}[Y] - \sqrt{\frac{\text{Var}[Y]}{\text{Var}[X]}} \text{E}[X] \right) + \sqrt{\frac{\text{Var}[Y]}{\text{Var}[X]}} x_0 = \left(\text{E}[Y] - \frac{1}{\gamma} \frac{\text{Cov}[Y, X]}{\text{Var}[X]} \text{E}[X] \right) + \frac{1}{\gamma} \frac{\text{Cov}[Y, X]}{\text{Var}[X]} x_0$$